Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

Perfomax **May 2026–present**

AI Advisor

National University of Singapore **2022–present**

Postdoc under Kostya Novoselov

Leadership AI4X 2025/6 conference · ICLR 2025 workshop

- Program Chair for the AI4X 2025/6 conferences.
- Main organizer of the ICLR 2025 workshop on multiscale machine learning.

Transformer for generating symmetric crystals ICML 2025 · code

- Created a generative model for material design based on symmetry inductive bias.
- Implemented the model in PyTorch.
- Achieved the best generation quality and diversity among space-group-conditioned models.
- Led a team of 6.

ML for predicting properties of defects in 2D crystals paper 1 · paper 2 · patent · conference · code

- Created a sparse representation of crystals with defects, achieving 4x energy prediction error reduction.
- Developed code for reproducible parallel running of machine learning experiments.
- Led a team of 3.

Constructor Group **February 2021–present**

Machine Learning Consultant

AI/ML Consulting

CERN [Yandex → HSE University] **2014–2022**

Researcher

2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.

Muon identification at the LHCb experiment at CERN paper · poster · IDAO-2019

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

CatBoost aka fighting biases with dynamic boosting paper

- Set up a distributed system for running experiments for the team with Bash and Python.
- Studied gradient boosting improvements with experiments on toy data.

Education

Higher School of Economics (HSE University) **2016–2020**
PhD in Computer Science, supervisor Andrey Ustyuzhanin; Machine Learning for particle identification in the LHCb detector.

Sapienza — Universita di Roma **2016–2020**
PhD in Physics (double degree with HSE), supervisor Barbara Sciascia.

Product Management course at Yandex **Spring 2018**
Focused coursework in product strategy and execution.

Yandex School of Data Analysis **2013–2015**
Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology
MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.
BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

Mentorship

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 2).
- Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
- Co-PI of a $3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage (1, 2), blog posts (1, 2), a science slam, interviews (1, 2), and a public talk.
- Taught at the Machine Learning in High Energy Physics Summer School in 2015, 2016, 2017, 2018, 2019, 2020, 2021, and 2023.
- Lecturer in the Coursera Advanced Machine Learning specialisation, which is award-winning.
- Taught and developed an introductory data analysis course at HSE.
- Delivered more than 20 conference talks.

Service

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Other

- Aspiring stand-up comedian; bombed more than 20 times at open mics.
- International Junior Science Olympiad 2008 in Korea, silver medal.

Skills

- **ML & AI** — Structured, non-text, domain-specific data
- **Python** — Research coding
- **PyTorch** — Deep learning
- **Linux** — Administration
- **Research Writing & Speaking**
- **Leadership & Mentorship**

Selected Publications

1. **Kazeev, Nikita**, and Phan Bui Nhat Huyen. Position: Align AI to Our Aspirations, Not Our Flaws. Pluralistic Alignment Workshop at ICML 2026, June 2026.
2. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Budennyy, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials. ICML 2026 AI for Science Workshop, May 2026.
3. Jose Recatala-Gomez, **Nikita Kazeev**, et al. Generative design of inorganic materials. arXiv preprint arXiv:2604.14082, April 2026.
4. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourrier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. LeMat-GenBench: Bridging the gap between crystal generation and materials discovery. AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
5. Stephen G. Dale, **Nikita Kazeev**, et al. AI4X Roadmap: Artificial Intelligence for the advancement of scientific pursuit and its future directions. arXiv preprint arXiv:2511.20976, November 2025.
6. **Kazeev, Nikita**, et al. WyckoffTransformer: Generation of Symmetric Crystals. ICML 2025, May 2025.
7. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. Sparse representation for machine learning the properties of defects in 2D materials. npj Comput Mater 9, 113 (2023).